

Data analysis with R

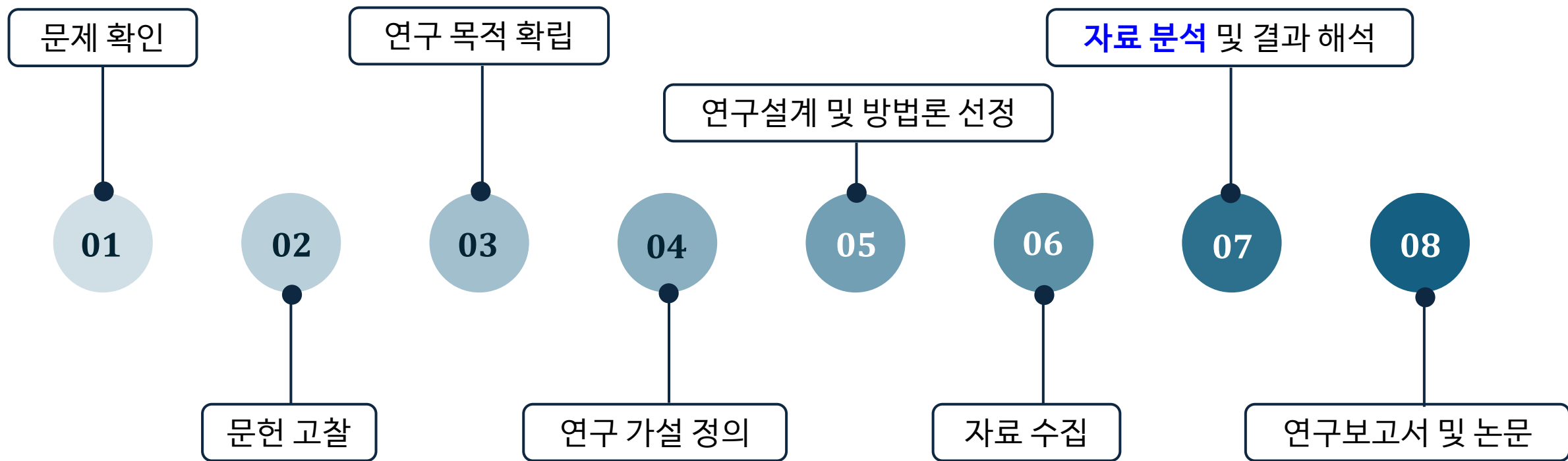
헬스케어데이터분석개론 13강

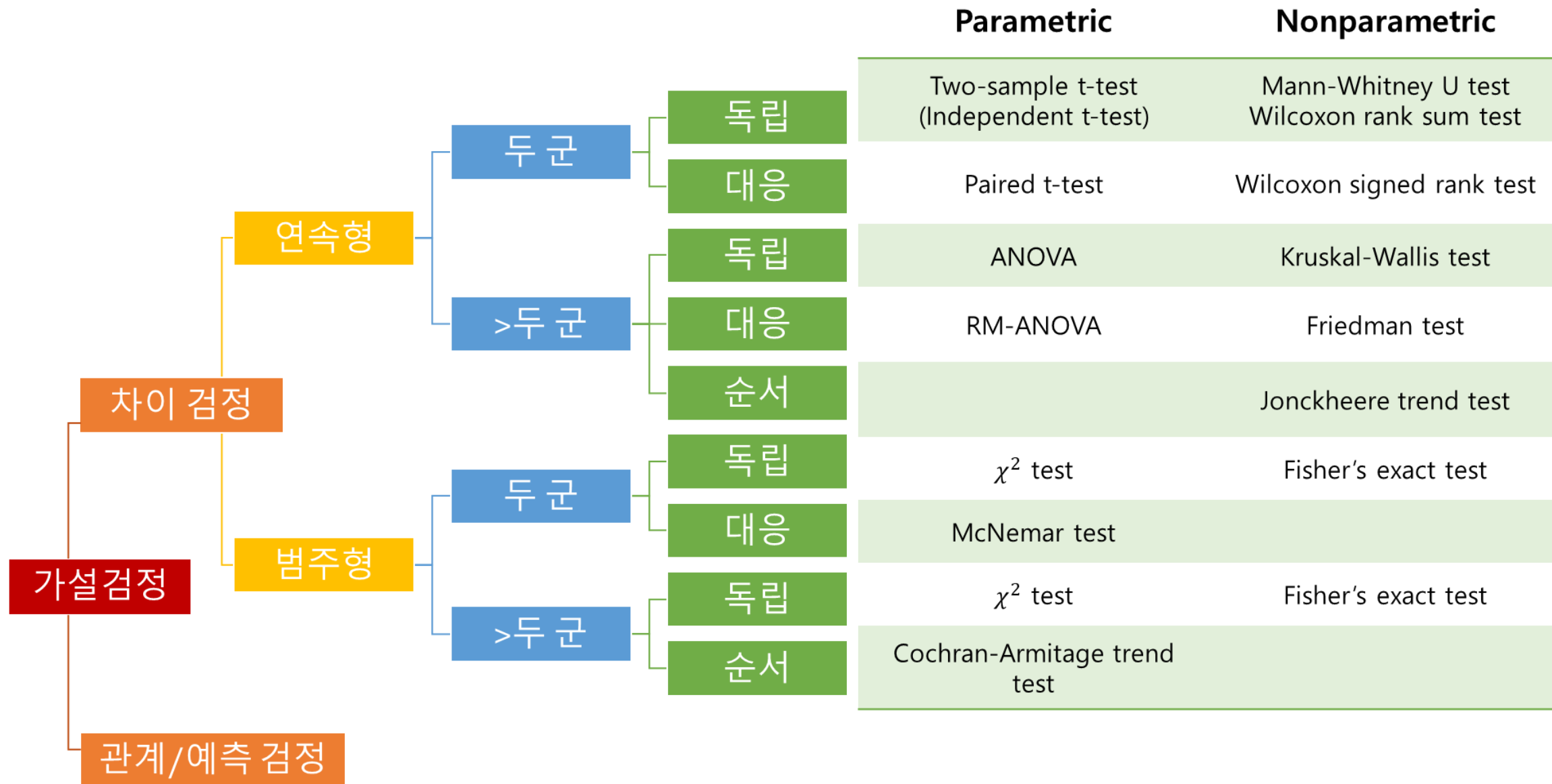
지상아

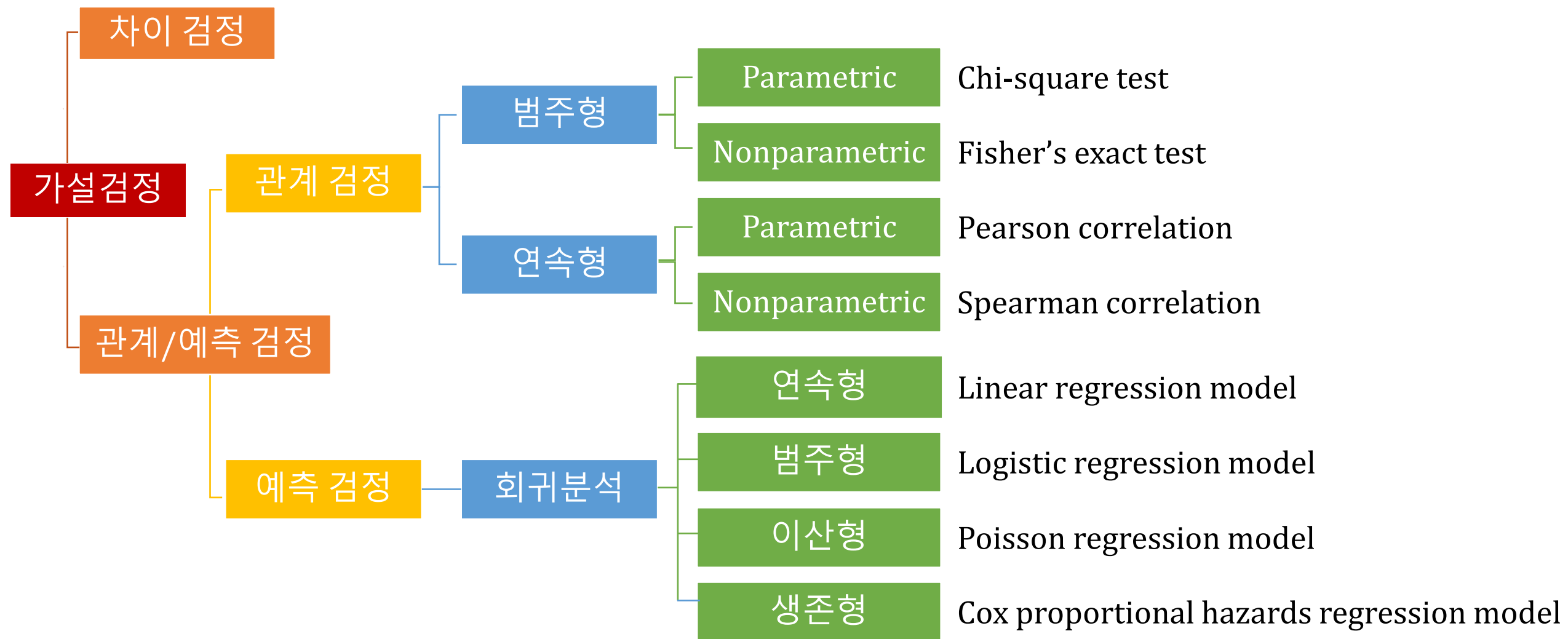
삼성서울병원 의학통계센터

sszzgg91@naver.com

To submit a paper ...







데이터 분석 실습

연구목적

CVD 발생에 영향을 미치는 인자는 무엇인가?

분석내용

- Table 1: Baseline characteristics for the study population
- Table 2: Comparison between those with CVD and those without CVD
- Figure 1: Kaplan-meier curve for CVD
- Table 3: Risk factor analysis for CVD

연구방법론

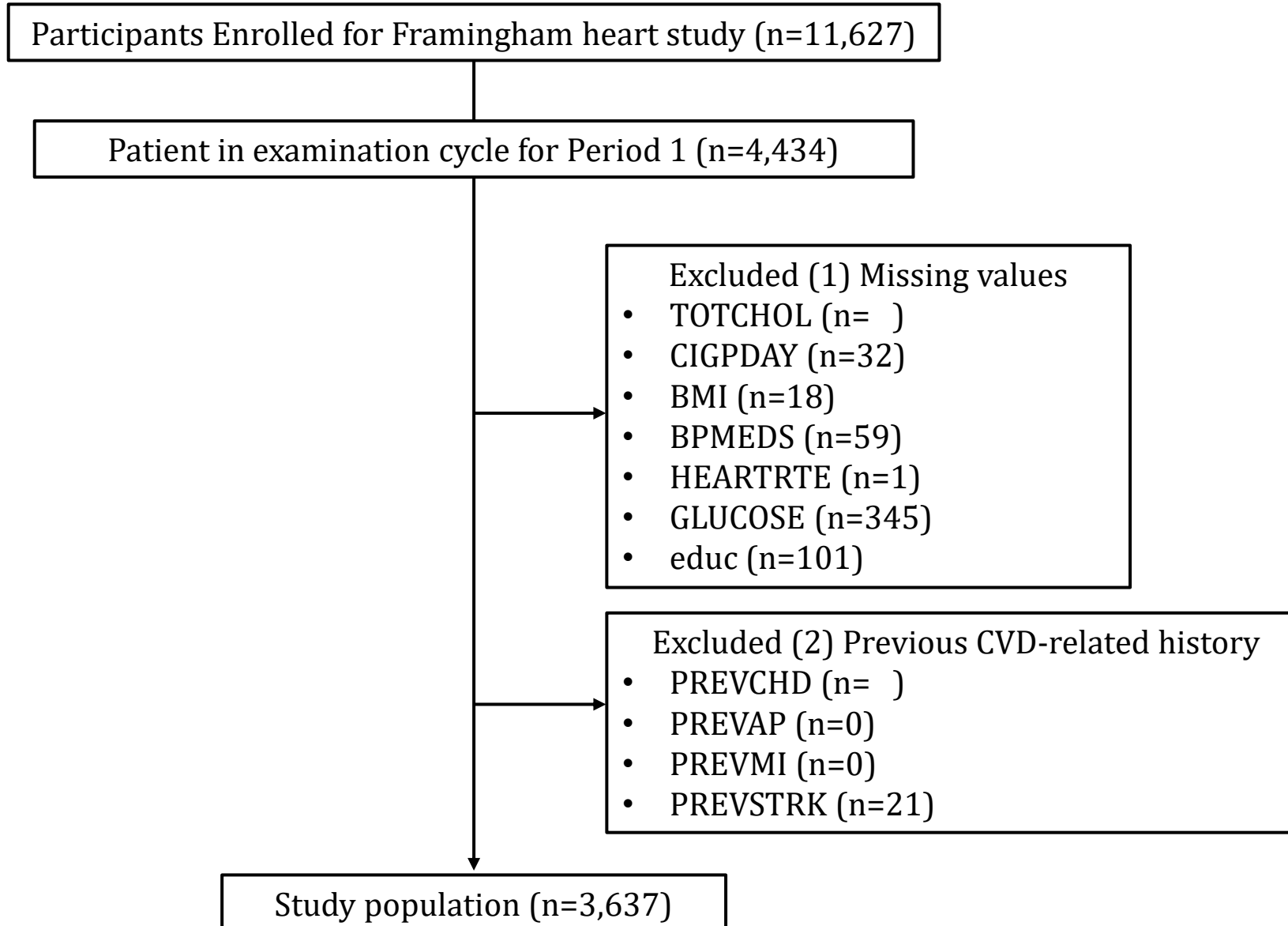
- Table 1: Summary Statistics for each characteristics
- Table 2
 - Two-sample t-test or Wilcoxon rank-sum test for continuous variables
 - Chi-square test or Fisher's exact test for categorical variables
- Figure 1: Survival probabilities using Kaplan-Meier method.
- Table 3:
 - Univariable analysis:
Hazard ratios (HRs) with 95% CI estimated from Cox's proportional hazard regression model.
 - Multivariable analysis:
Variables with a P-value less than 0.1 in the univariable model were included in the multivariable analysis.

수집된 자료: Framingham Heart Study

The Framingham Heart Study is a long term prospective study of the etiology of cardiovascular disease among a population of free living subjects in the community of Framingham, Massachusetts.

No.	변수명	설명
1	RANDID	개인식별번호
2	SEX	성별, (1: 남자, 2: 여자)
3	TOTCHOL	총콜레스테롤(total cholesterol) (mg/dl)
4	AGE	조사 당시 나이
5	SYSBP	수축기혈압(systolic blood pressure, SBP) (mmHg)
6	DIABP	이완기혈압(diastolic blood pressure, DBP) (mmHg)
7	CURSMOKE	흡연여부, (0: 현재 피우지 않음, 1: 현재 피움)
8	BMI	체질량지수(body mass index)
9	DIABETES	당뇨 여부 (casual glucose level로 측정) (0: Normal, 1: Diabetes)
10	BPMEDS	항고혈압 약 복용 여부 (0: No, 1: Yes)
11	HEARTRTE	심박수 (beats/min)
12	PREVHYP	과거 Hypertension 진단 여부 (0: No, 1: Yes)
13	PREVCHD	과거 Coronary heart disease (CHD) 진단 여부 (0: No, 1: Yes)
14	PREVAP	과거 Angina pectoris (AP) 진단 여부 (0: No, 1: Yes)
15	PREVMI	과거 Myocardial Infarction (MI) 진단 여부 (0: No, 1: Yes)
16	PREVSTRK	과거 Stroke 진단 여부 (0: No, 1: Yes)
17	PERIOD	조사 기간 (Examination cycle) 1 = Period 1 (n = 4434), 2 = Period 2 (n = 3930), 3 = Period 3 (n = 3263).
18	CVD	추적 관찰 (Follow-up) 이후 CVD 발생 여부 (0: No, 1: Yes)
19	TIMECVD	CVD 발생했을 때의 시간

Flow Chart for the study population



Flow Chart for the study population

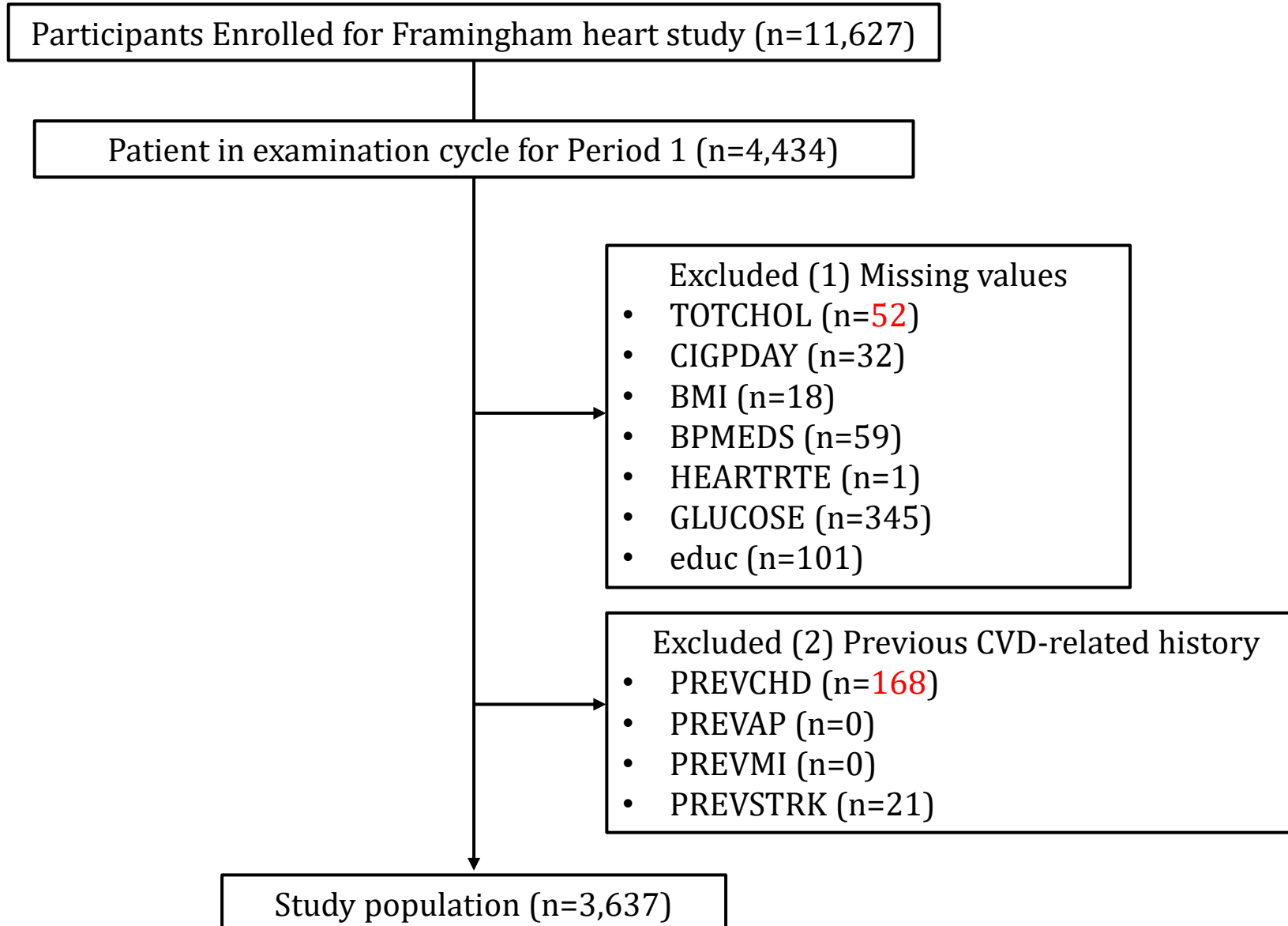


Table 1: Baseline characteristics for the study population

	Total (N= 3,637)
Sex (SEX)	
- Male	
- Female	
Age (AGE)	49.0 (42.0, 56.0)
Body mass index (BMI)	25.4 (23.1, 28.0)
Systolic blood pressure (SYSBP)	128.0 (117.0, 143.5)
Heart rate (HEARTRTE)	
Total cholesterol (TOTCHOL)	
Diabetes (DIABETES)	
Current smoking (CURSMOKE)	1,784 (49.1%)
Hypertension (PREVHYP)	1,125 (30.9%)
Anti-hypertensive medication use (BPMEDS)	105 (2.9%)

- Categorical variables
 - Frequency (percentage)
- Continuous variables
 - Normality assumption (Shapiro-Wilk test)
 - Not violated
 - Mean ± SD
 - Violated
 - Median (Q1, Q3)

Table 1: Baseline characteristics for the study population

11

	Total (N= 3,637)
Sex (SEX)	
- Male	1,614 (44.4%)
- Female	2,023 (55.6%)
Age (AGE)	49.0 (42.0, 56.0)
Body mass index (BMI)	25.4 (23.1, 28.0)
Systolic blood pressure (SYSBP)	128.0 (117.0, 143.5)
Heart rate (HEARTRTE)	75.0 (68.0, 82.0)
Total cholesterol (TOTCHOL)	234.0 (206.0, 263.0)
Diabetes (DIABETES)	98 (2.7%)
Current smoking (CURSMOKE)	1,784 (49.1%)
Hypertension (PREVHYP)	1,125 (30.9%)
Anti-hypertensive medication use (BPMEDS)	105 (2.9%)

- Categorical variables
 - Frequency (percentage)
- Continuous variables
 - Normality assumption (Shapiro-Wilk test)
 - Not violated
 - Mean $\pm$ SD
 - Violated
 - Median (Q1, Q3)

Table 2: Comparison between those with CVD and those without CVD

	No CVD (N= 2,763)	CVD (N= 874)	P-value
Sex (SEX)			
- Male			
- Female			
Age (AGE)	47.0 (41.0, 55.0)	53.0 (46.0, 60.0)	<.0001
Body mass index (BMI)	25.1 (22.9, 27.6)	26.4 (24.0, 28.9)	<.0001
Systolic blood pressure (SYSBP)	126.0 (115.0, 140.0)	137.0 (125.0, 154.4)	<.0001
Heart rate (HEARTRTE)			
Total cholesterol (TOTCHOL)			
Diabetes (DIABETES)			
Current smoking (CURSMOKE)	1,328 (48.1%)	456 (52.2%)	0.0376
Hypertension (PREVHYP)	701 (25.4%)	424 (48.5%)	<.0001
Anti-hypertensive medication use (BPMEDS)	59 (2.1%)	46 (5.3%)	<.0001

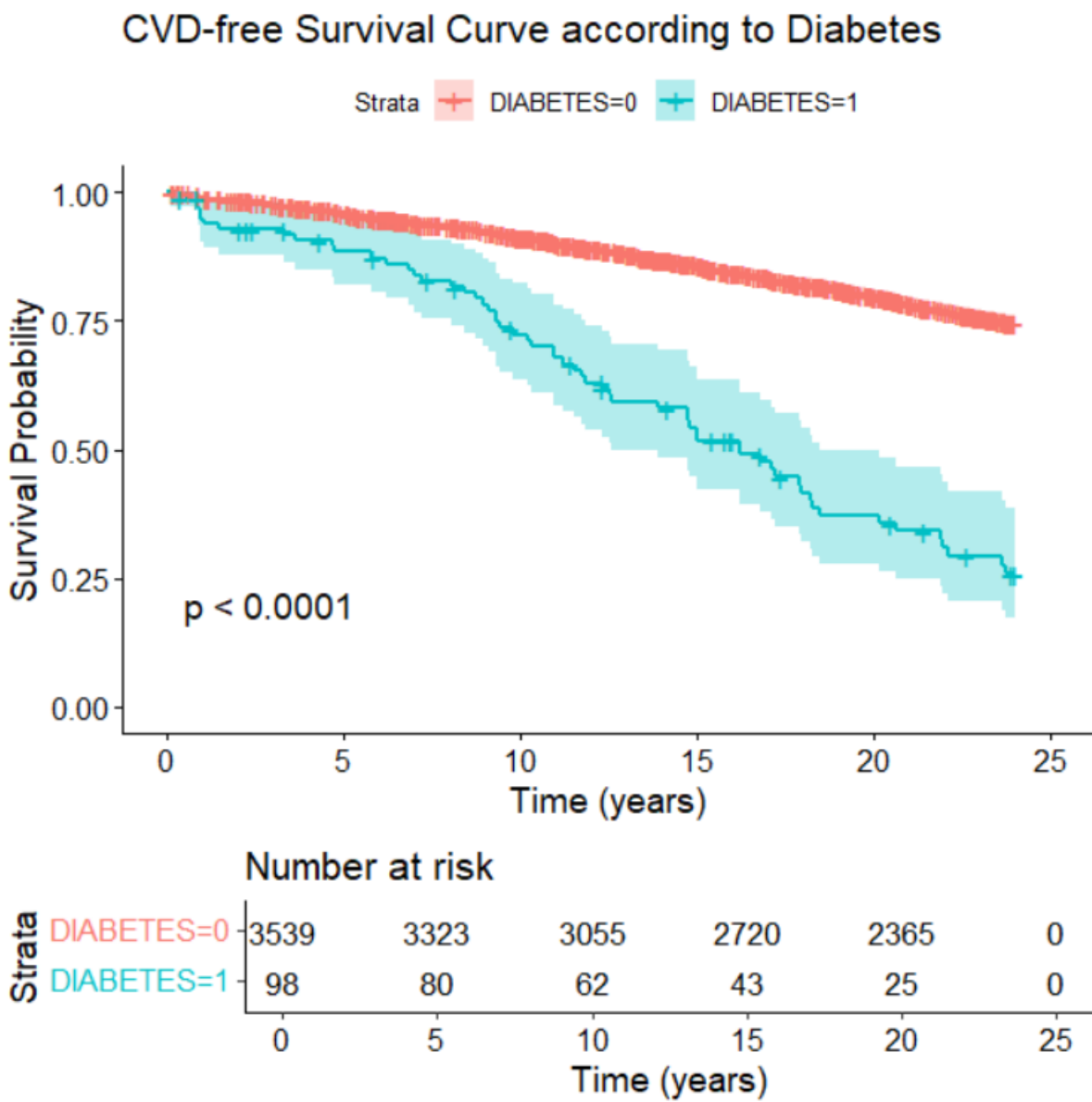
- Two-sample t-test or Wilcoxon rank-sum test for continuous variables
- Chi-square test or Fisher's exact test for categorical variables

Table 2: Comparison between those with CVD and those without CVD

	No CVD (N= 2,763)	CVD (N= 874)	P-value
Sex (SEX)			<.0001
- Male	1,100 (39.8%)	514 (58.8%)	
- Female	1,663 (60.2%)	360 (41.2%)	
Age (AGE)	47.0 (41.0, 55.0)	53.0 (46.0, 60.0)	<.0001
Body mass index (BMI)	25.1 (22.9, 27.6)	26.4 (24.0, 28.9)	<.0001
Systolic blood pressure (SYSBP)	126.0 (115.0, 140.0)	137.0 (125.0, 154.4)	<.0001
Heart rate (HEARTRTE)	75.0 (68.0, 83.0)	75.0 (68.0, 82.0)	0.5273
Total cholesterol (TOTCHOL)	231.0 (203.0, 261.0)	242.0 (215.0, 272.0)	<.0001
Diabetes (DIABETES)	39 (1.4%)	59 (6.8%)	<.0001
Current smoking (CURSMOKE)	1,328 (48.1%)	456 (52.2%)	0.0376
Hypertension (PREVHYP)	701 (25.4%)	424 (48.5%)	<.0001
Anti-hypertensive medication use (BPMEDS)	59 (2.1%)	46 (5.3%)	<.0001

- Two-sample t-test or Wilcoxon rank-sum test for continuous variables
- Chi-square test or Fisher's exact test for categorical variables

Figure 1: Kaplan-meier curve for CVD



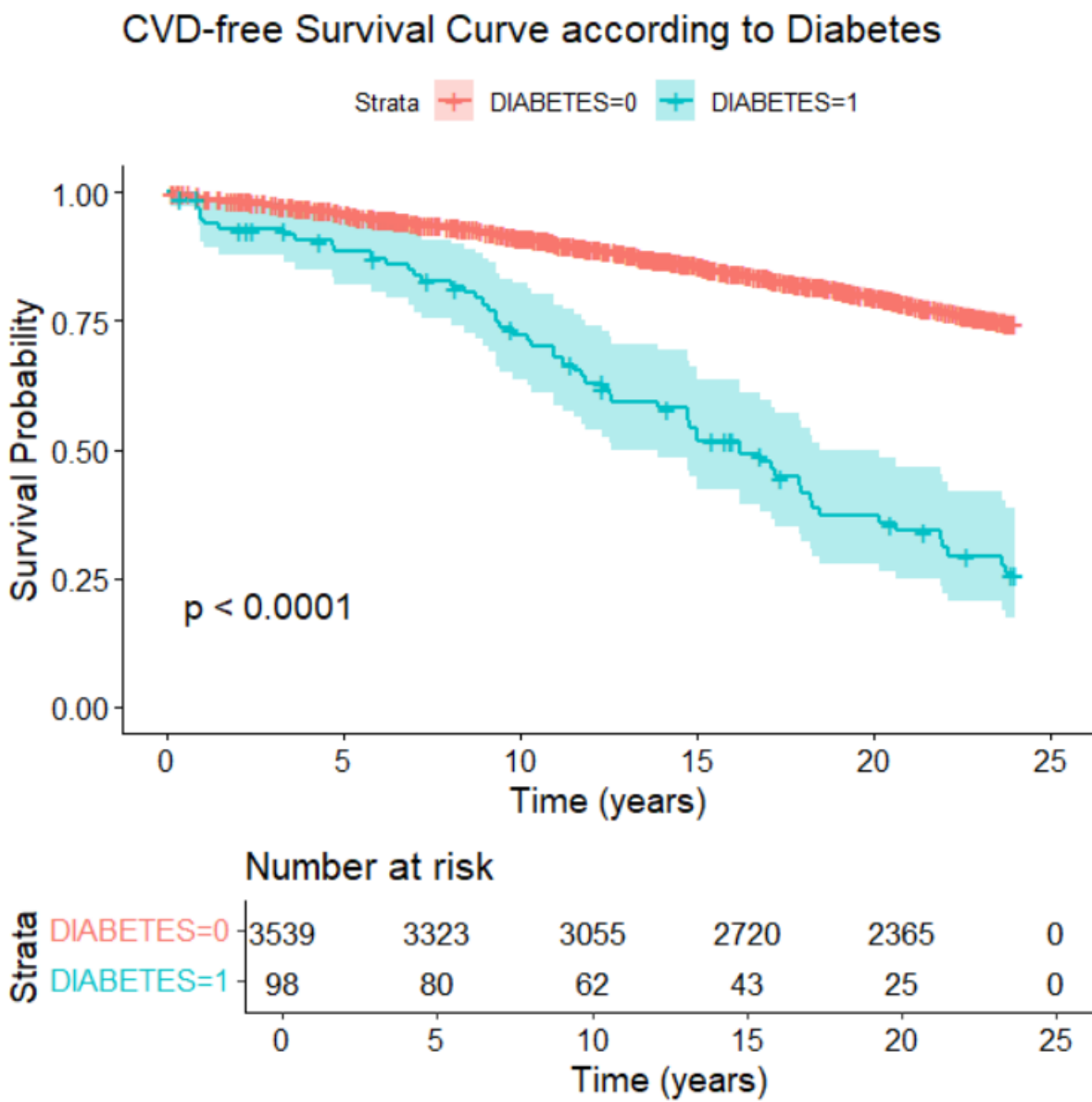
- Survival probabilities using Kaplan–Meier method.

DIABETES=0					
years	n.risk	n.event	survival	95% LCL	95% UCL
5	3323	141		0.953	0.966
10	3055	159		0.904	0.923
15	2720	174		0.848	0.871
20	2365	188		0.785	0.812

DIABETES=1					
years	n.risk	n.event	survival	95% LCL	95% UCL
5	80	11		0.821	0.951
10	62	14		0.637	0.823
15	43	16		0.434	0.648
20	25	11		0.278	0.498

Log-rank P-value=

Figure 1: Kaplan-meier curve for CVD



- Survival probabilities using Kaplan–Meier method.

DIABETES=0					
years	n.risk	n.event	survival	95% LCL	95% UCL
5	3323	141	0.960	0.953	0.966
10	3055	159	0.913	0.904	0.923
15	2720	174	0.860	0.848	0.871
20	2365	188	0.798	0.785	0.812

DIABETES=1					
years	n.risk	n.event	survival	95% LCL	95% UCL
5	80	11	0.883	0.821	0.951
10	62	14	0.724	0.637	0.823
15	43	16	0.530	0.434	0.648
20	25	11	0.372	0.278	0.498

Log-rank P-value <.0001

Table 3: Risk factor analysis for CVD

	Univariable model			Multivariable model		
	HR	95% CI	P-value	HR	95% CI	P-value
Sex (SEX)						
- Male						
- Female						
Age (AGE)	1.068	(1.060, 1.077)	<.0001	1.055	(1.046, 1.064)	<.0001
Body mass index (BMI)	1.072	(1.057, 1.087)	<.0001	1.029	(1.012, 1.046)	0.0008
Systolic blood pressure (SYSBP)	1.023	(1.021, 1.026)	<.0001	1.013	(1.009, 1.017)	<.0001
Heart rate (HEARTRTE)						
Total cholesterol (TOTCHOL)						
- Less than 200						
- Equal or more than 200						
Diabetes (DIABETES)						
Current smoking (CURSMOKE)	1.163	(1.019, 1.329)	0.0254	1.411	(1.225, 1.626)	<.0001
Hypertension (PREVHYP)	2.739	(2.397, 3.128)	<.0001	1.347	(1.123, 1.616)	0.0013
Anti-hypertensive medication use (BPMEDS)	2.816	(2.091, 3.792)	<.0001	1.353	(0.989, 1.851)	0.0582

- Did all variables meet proportional hazards assumption?
- Was there severe level of multicollinearity among covariates?

Table 3: Risk factor analysis for CVD

	Univariable model			Multivariable model		
	HR	95% CI	P-value	HR	95% CI	P-value
Sex (SEX)			<.0001			
- Male	Ref					
- Female	0.489	(0.427, 0.559)				
Age (AGE)	1.068	(1.060, 1.077)	<.0001	1.055	(1.046, 1.064)	<.0001
Body mass index (BMI)	1.072	(1.057, 1.087)	<.0001	1.029	(1.012, 1.046)	0.0008
Systolic blood pressure (SYSBP)	1.023	(1.021, 1.026)	<.0001	1.013	(1.009, 1.017)	<.0001
Heart rate (HEARTRTE)	1.000	(0.995, 0.006)	0.9347			
Total cholesterol (TOTCHOL)			<.0001			
- Less than 200	Ref					
- Equal or more than 200	1.817	(1.489, 2.218)				
Diabetes (DIABETES)	4.220	(3.237, 5.503)	<.0001			
Current smoking (CURSMOKE)	1.163	(1.019, 1.329)	0.0254	1.411	(1.225, 1.626)	<.0001
Hypertension (PREVHYP)	2.739	(2.397, 3.128)	<.0001	1.347	(1.123, 1.616)	0.0013
Anti-hypertensive medication use (BPMEDS)	2.816	(2.091, 3.792)	<.0001	1.353	(0.989, 1.851)	0.0582

- Did all variables meet proportional hazards assumption?
- Was there severe level of multicollinearity among covariates?

Table 3: Risk factor analysis for CVD

	Univariable model			Multivariable model		
	HR	95% CI	P-value	HR	95% CI	P-value
Sex (SEX)			<.0001			<.0001
- Male	Ref			Ref		
- Female	0.489	(0.427, 0.559)		0.430	(0.373, 0.496)	
Age (AGE)	1.068	(1.060, 1.077)	<.0001	1.055	(1.046, 1.064)	<.0001
Body mass index (BMI)	1.072	(1.057, 1.087)	<.0001	1.029	(1.012, 1.046)	0.0008
Systolic blood pressure (SYSBP)	1.023	(1.021, 1.026)	<.0001	1.013	(1.009, 1.017)	<.0001
Heart rate (HEARTRTE)	1.000	(0.995, 0.006)	0.9347			
Total cholesterol (TOTCHOL)			<.0001			0.0023
- Less than 200	Ref			Ref		
- Equal or more than 200	1.817	(1.489, 2.218)		1.367	(1.118, 1.672)	
Diabetes (DIABETES)	4.220	(3.237, 5.503)	<.0001	2.646	(2.019, 3.466)	<.0001
Current smoking (CURSMOKE)	1.163	(1.019, 1.329)	0.0254	1.411	(1.225, 1.626)	<.0001
Hypertension (PREVHYP)	2.739	(2.397, 3.128)	<.0001	1.347	(1.123, 1.616)	0.0013
Anti-hypertensive medication use (BPMEDS)	2.816	(2.091, 3.792)	<.0001	1.353	(0.989, 1.851)	0.0582

- Did all variables meet proportional hazards assumption?
- Was there severe level of multicollinearity among covariates?

	P-value
Sex (SEX)	
- Male	
- Female	
Age (AGE)	0.2625
Body mass index (BMI)	0.1303
Systolic blood pressure (SYSBP)	0.1179
Heart rate (HEARTRTE)	
Total cholesterol (TOTCHOL)	
- Less than 200	
- Equal or more than 200	
Diabetes (DIABETES)	
Current smoking (CURSMOKE)	0.9328
Hypertension (PREVHYP)	0.0532
Anti-hypertensive medication use (BPMEDS)	0.4077

- Did all variables meet proportional hazards assumption?
- Was there severe level of multicollinearity among covariates?

참고: Shoenfeld residual test for PH assumption

	P-value
Sex (SEX)	0.2420
- Male	
- Female	
Age (AGE)	0.2625
Body mass index (BMI)	0.1303
Systolic blood pressure (SYSBP)	0.1179
Heart rate (HEARTRTE)	0.4005
Total cholesterol (TOTCHOL)	0.4206
- Less than 200	
- Equal or more than 200	
Diabetes (DIABETES)	0.1772
Current smoking (CURSMOKE)	0.9328
Hypertension (PREVHYP)	0.0532
Anti-hypertensive medication use (BPMEDS)	0.4077

- Did all variables meet proportional hazards assumption?
- Was there severe level of multicollinearity among covariates?

- Preprocessing data
 - Remove outlier
 - Transformation (ex. Logarithmic, Box-cox)
 - Categorization for continuous variable
- Time-dependent covariates
 - Interaction with time
- Stratified cox regression model for categorical variable
- Remove the covariate

Supplementary Table 1: Risk factor analysis for CVD (Yes or No)

22

	Univariable model			Multivariable model		
	OR	95% CI	P-value	OR	95% CI	P-value
Sex (SEX)						
- Male						
- Female						
Age (AGE)	1.064	(1.054, 1.074)	<.0001	1.049	(1.038, 1.060)	<.0001
Body mass index (BMI)	1.084	(1.064, 1.104)	<.0001	1.040	(1.018, 1.063)	0.0003
Systolic blood pressure (SYSBP)	1.024	(1.021, 1.027)	<.0001	1.012	(1.007, 1.017)	<.0001
Heart rate (HEARTRTE)						
Total cholesterol (TOTCHOL)						
- Less than 200						
- Equal or more than 200						
Diabetes (DIABETES)						
Current smoking (CURSMOKE)	1.179	(1.012, 1.373)	0.0342	1.459	(1.224, 1.739)	<.0001
Hypertension (PREVHYP)	2.772	(2.367, 3.246)	<.0001	1.411	(1.121, 1.776)	0.0034
Anti-hypertensive medication use (BPMEDS)	2.546	(1.718, 3.773)	<.0001	1.118	(0.720, 1.730)	0.6242

Supplementary Table 1: Risk factor analysis for CVD (Yes or No)

	Univariable model			Multivariable model		
	OR	95% CI	P-value	OR	95% CI	P-value
Sex (SEX)			<.0001			
- Male	Ref					
- Female	0.463	(0.397, 0.541)	<.0001			
Age (AGE)	1.064	(1.054, 1.074)	<.0001	1.049	(1.038, 1.060)	<.0001
Body mass index (BMI)	1.084	(1.064, 1.104)	<.0001	1.040	(1.018, 1.063)	0.0003
Systolic blood pressure (SYSBP)	1.024	(1.021, 1.027)	<.0001	1.012	(1.007, 1.017)	<.0001
Heart rate (HEARTRTE)	0.999	(0.992, 1.005)	0.6406			
Total cholesterol (TOTCHOL)			<.0001			
- Less than 200	Ref					
- Equal or more than 200	1.927	(1.549, 2.398)				
Diabetes (DIABETES)	5.056	(3.349, 7.634)	<.0001			
Current smoking (CURSMOKE)	1.179	(1.012, 1.373)	0.0342	1.459	(1.224, 1.739)	<.0001
Hypertension (PREVHYP)	2.772	(2.367, 3.246)	<.0001	1.411	(1.121, 1.776)	0.0034
Anti-hypertensive medication use (BPMEDS)	2.546	(1.718, 3.773)	<.0001	1.118	(0.720, 1.730)	0.6242

Supplementary Table 1: Risk factor analysis for CVD (Yes or No)

24

	Univariable model			Multivariable model		
	OR	95% CI	P-value	OR	95% CI	P-value
Sex (SEX)			<.0001			<.0001
- Male	Ref			Ref		
- Female	0.463	(0.397, 0.541)	<.0001	0.426	(0.358, 0.506)	
Age (AGE)	1.064	(1.054, 1.074)	<.0001	1.049	(1.038, 1.060)	<.0001
Body mass index (BMI)	1.084	(1.064, 1.104)	<.0001	1.040	(1.018, 1.063)	0.0003
Systolic blood pressure (SYSBP)	1.024	(1.021, 1.027)	<.0001	1.012	(1.007, 1.017)	<.0001
Heart rate (HEARTRTE)	0.999	(0.992, 1.005)	0.6406			
Total cholesterol (TOTCHOL)			<.0001			0.0012
- Less than 200	Ref			Ref		
- Equal or more than 200	1.927	(1.549, 2.398)		1.483	(1.169, 1.882)	
Diabetes (DIABETES)	5.056	(3.349, 7.634)	<.0001	3.431	(2.210, 5.328)	<.0001
Current smoking (CURSMOKE)	1.179	(1.012, 1.373)	0.0342	1.459	(1.224, 1.739)	<.0001
Hypertension (PREVHYP)	2.772	(2.367, 3.246)	<.0001	1.411	(1.121, 1.776)	0.0034
Anti-hypertensive medication use (BPMEDS)	2.546	(1.718, 3.773)	<.0001	1.118	(0.720, 1.730)	0.6242

If you have ANY question,
please contact to
sszzgg91@naver.com